MILITARY SPECIFICATION
PROJECTILE HE, 37MM, M63
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Projectile HE, 37MM, M63, Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-P-10025	-	Packaging, Packing and Marking for Inter Plant Shipment of Inert Ammunition Components
MIL-I-45208	-	Inspection Requirements, General Specification for

STANDARDS

Federal
Federal Test Methods
Standard No. 151 - Metal Test Methods

Military

MIL-STD-10	-	Surface Roughness, Waviness and Lay
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Inspection Terms and Definitions
MIL-STD-639	-	Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

Ordnance Corps

F8597556	-	Projectile, HE, 37MM, M63, Metal Parts Assembly

FSC 1310

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F7548319 - Carton Ammunition for Inter Plant Shipment of Projectile Empty
IEL-8597556 - Index of Inspection Equipment List for Projectile HE, 37MM, M63

3. REQUIREMENTS

3.1 General.- Material, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specification and the following:

3.2 Initial Production Sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Heat Treatment or Stress Relief.- Heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in any component subsequently selected for test, the mechanical properties specified on the applicable component drawing.

3.4 Mechanical Properties.- Mechanical properties shall conform to all the requirements as stated in the applicable specification or on the applicable drawing.

3.4.1 Rotating Band.- Upon removal from the projectile, the rotating band shall have the following mechanical properties; tensile strength 32,000 pounds per square inch and a minimum elongation of 25 per cent per inch of gage length.

3.4.2 Band Seating.- The average diametrical clearance between the rotating band and band seat shall not exceed .005 inches.

3.5 Surface Treatments and Coatings.- Surface treatments and coatings methods and materials shall be such as to comply with requirements of the specifications listed on the applicable drawing.

3.6 Functional

3.6.1 Security.- There shall be no loss or breakup of any component either in the gunbore or in flight.

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3.7 Workmanship.- The requirements for workmanship shall be in accordance with applicable component drawings and the following.

3.7.1 Metal Defect.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity and any similar defects.

3.7.2 Burr.- No component shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign Matter.- All parts and assemblies shall be free of chips, dirt, grease, rust corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor Inspection.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and test shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection Terms and Definitions.- Reference shall be made to Standard MIL-STD-109 to define the inspection terms used.

4.1.2 Inspection.- Inspection shall be in accordance with Specifications MIL-A-2550 and MIL-I-45208, except as specified herein.

4.2 Initial Production Sample.- Prior to the initiation of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 25 Metal Part assemblies complete including paint; 5 metal part assemblies, less paint. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of Specification MIL-I-45208. After inspection and provisional acceptance at source, the initial production

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sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection Provisions

4.3.1 Lot

4.3.1.1 Submittal of Product.- The contractor shall submit the product in accordance with Standard MIL-STD-105.

4.3.1.2 Size of Lot.- For purposes of identification and shipment, the lot size shall be as agreed upon between the contractor and the contracting officer, provided the homogeneity of the inspection lot has been retained.

4.3.1.3 Forging Lot.- Forging lots shall not contain more than one heat of steel.

4.3.1.4 Miscellaneous Lots.- Small lots (such as abnormally small heats of components which have become separated from previously delivered lots, etc.) may at the discretion of the manufacture, be grouped into miscellaneous forging lots. Such lot shall be identified by a miscellaneous lot code number.

4.3.1.5 Grouping.- Heat treated, stress relieved or miscellaneous lots shall remain grouped together for lot identity purposes, according to forging or stress relieved lots of each group, until such time as the results of the tensile test indicated that the lot has complied with the mechanical properties requirement of the applicable drawing.

4.3.2 Examination.- Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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4.3.2.1 Body, prior to banding, part no. 8597557 covering a detail on Drawing F8597556.

Defect	Method of Inspection
CRITICAL	
None Defined	
MAJOR	
101. Diameter of band seat	Gage
102. Width of band seat	Gage
103. Location of band seat	Gage
MINOR	
201. Foreign Matter	Visual

4.3.2.2 Body, Metal Part Assembly, prior to painting Drawing F8597556.

CRITICAL	
Metal defective	
MAJOR	
101. Concentricity of bourrelet with charge cavity	Gage
102. Concentricity of bourrelet with shell base	Gage
103. Concentricity of bourrelet with ogive	Gage
104. Concentricity of pitch diameter of thread with charge cavity	Gage
105. Concentricity of rotating band with bourrelet	Gage
106. Concentricity of pitch diameter of thread with bourrelet	Gage
107. Diameter of rotating band, min.	Gage
108. Diameter of body cavity under rotating band, max.	Gage

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Defect	Method of Inspection
MAJOR (Cont'd)	
109. Diameter of bourrelet, min.	Gage
110. Depth of crimping groove	Gage
111. Location of crimping groove	Gage
112. Profile of crimping groove	Gage
MINOR	
201. Diameter of undercut, front and rear of band	Gage
202. Diameter of rotating band, max.	Gage
203. Diameter of body cavity under rotating band, min.	Gage
204. Diameter of body	Gage
205. Diameter of relief groove in rotating band	Gage
206. Diameter of fuze thread counterbore	Gage
207. Depth of cavity	Gage
208. Depth of fuze thread counterbore	Gage
209. Total length	Gage
210. Length from rear end to internal datum	Gage
211. Length to external datum diameter	Gage
212. Width and profile of rotating band	Gage
213. Width and profile of relief groove in rotating band	Gage
214. Location of rotating band	Gage
215. Surface finish, improper	Visual
216. Burr	Visual
217. Foreign Matter	Visual

4.3.2.3 Metal Part Assembly, Drawing F8597556

CRITICAL	
1. Pool of paint in nose of cavity	Visual
MAJOR	
101. Pitch diameter of fuze thread, max.	Gage
102. Minor diameter of fuze thread, max.	Gage
103. Diameter of bourrelet, max.	Gage
104. Length of fuze thread, min.	Gage
105. Thread damaged or not full	Visual

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Defect	Method of Inspection
MINOR	
201. Pitch diameter of fuze thread, min.	Gage
202. Length of fuze thread, max.	Gage
203. Weight	Gage
204. Interior coating not dry	Tactile
205. Exterior paint not dry	Tactile
206. Rotating band damaged	Visual
207. Interior coating inadequate or defective	Visual
208. Exterior coating inadequate or defective	Visual
209. Marking, missing, incorrect or illegible	Visual
210. Paint on rotating band	Visual
211. Burr	Visual
212. Foreign Matter	Visual

4.3.2.4 Packed Carton

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Support missing or damaged	Visual
202. Separator missing or damaged	Visual
203. Improper assembly	Visual
204. Carton damaged, cut or gouged through all layers	Visual
205. Contents loose	Visual
206. Carton not completely sealed	Visual
207. Marking missing, incorrect or illegible	Visual

4.3.2.5 Pallet

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Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Strapping missing loose or broken	Manual/Visual
202. Pallet damaged	Visual
203. Improperly packed	Visual

4.3.3 Inspection Testing.- The following tests shall be performed in accordance with provisions of 4.4 except as provided in Specification MIL-I-45208.

4.3.3.1 Mechanical Properties.-

4.3.3.1.1 Hardness.- One projectile body, after heat treatment or stress relief but prior to finish machining, shall be selected from each batch, or one body from each one hours production of a continuously operated conveyor type furnace shall be subjected to this test. Failure of the body to comply with the hardness requirements of the applicable drawing shall be cause for rejection of the batch or group represented by the sample.

4.3.3.1.1.1 Reheat Treatment or Restress Relief.- Batches or groups of projectile that have failed the hardness test may be reheat treated or restress relieved and shall be identified in such a manner that all projectile bodies contained therein can be identified through all subsequent operations including banding.

4.3.3.1.1.2 Resubmission.- Upon resubmission of a reheat treated or restress relieved batch or group, 2 projectile bodies shall be selected from each batch or group and subjected to the hardness test. Failure of either sample to comply with the hardness requirements of the applicable drawing shall be cause for rejection of all bodies represented by the sample. One body assembly from each reheat treated on restress relieved batch or group that has passed the hardness test shall, after banding, be subjected to the tension test as detailed in 4.3.3.1.2.

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4.3.3.1.2 Tension Test.- One body assembly, from each heat of steel and from each resubmitted batch or group shall be subjected to this test to determine compliance with the yield strength and elongation requirements of the applicable drawing. Failure of any test specimen to comply with the requirements shall be cause for rejection of the assemblies represented by the specimen.

4.3.3.2 Rotating Band.- One body assembly shall be selected for each 8 hours production or fraction thereof from the product of each banding machine in operation and subjected to the tension test. Failure of the band to comply with 3.4.1 shall be cause for rejection of the assemblies represented by the sample.

4.3.3.3 Band Seating.

4.3.3.3.1 Destructive Method.- When the destructive method of testing is employed, one body assembly shall be selected from each 8 hours production or fraction thereof per banding machine in operation. Failure of any band to comply with 3.4.2 shall be cause for rejection of the production represented by the sample.

4.3.3.4 Protective Coating.- The sample size and acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

4.3.3.5 Functional.- The following functional (proving ground) test will be performed by the Government:

4.3.3.5.1 Sample for functional testing will be fired in the phases indicated by Table I.

TABLE I

Test Phase	Sample Size	
	Preproduction	Production
Security	10	*10
Velocity-Accuracy	10	10

*After 15 successive samples (an accumulative sample of 150 projectiles) have been tested for metal parts security, without evidence of noncompliance with 3.6.1 the sample size will be reduced to 5 projectiles per lot, for an additional 10 lots (a total accumulative sample of 200 projectiles) without evidence of noncompliance, the sample size may be further reduced

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to 5 samples from each alternate lot. In the event of failure of a sample to comply with 3.6.1, the sample on lots following the failing lot shall revert to the original sample size. Returning to reduced sampling will be permitted only when each of the above steps have been complied with.

4.3.3.5.2 Security.- The samples selected in accordance with Table I will be subject to this test. Failure of any assembly of the production lot sample to comply with 3.6.1 will be cause for rejection of the lot.

4.3.3.5.3 Velocity - Accuracy.- The samples selected in accordance with Table I will be subjected to this test, for informational purposes only. Accuracy shall be measured in terms of maximum spread in inches, and probable error (PE) in mils. Velocity and pressure of each round fired will also be recorded.

4.3.4 Inspection Equipment.- The examination and tests shall be made using equipment listed on IEL-8597556, except as provided in Specification MIL-I-45208.

4.4 Test Methods and Procedures.

4.4.1 Mechanical Properties.

4.4.1.1 Hardness.- The hardness of the exterior surface of the projectile body, to the rear of the rotating band seat, shall be determined using a hardness tester or any other hardness testing equipment specified in 4.3.4. Five hardness readings shall be taken (approximately equidistant) on the circumference of each sample body. This test shall be performed using the criteria prescribed by Federal Test Method Standard NO. 151.

4.4.1.2 Tension Test.

4.4.1.2.1 Yield Strength and Elongation.- From the sample selected in accordance with 4.3.2.2. Two tension test specimens conforming as closely as possible to the largest obtainable type "R" specimen as prescribed by Federal Test Methods Standard No. 151 shall be taken longitudinally from opposite sides of the wall of each sample component. The criteria for determining yield strength and elongation shall be as prescribed by Federal Test Methods Standard No. 151.

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4.4.1.2.1.1 Replacement Specimen.- When failure of any test specimen can be attributed to a defective specimen or improper method of testing, the failing bar may be replaced by taking an additional test specimen from the same projectile body adjacent to the failed bar, or at the option of the contractor, two sample projectile from the same lot may be selected and tested as an original test.

4.4.2 Rotating Band Tension Test.- The rotating band selected in accordance with 4.3.3.2 shall be tested as follows: The test specimen shall be cut along the flattened length and prepared in such a manner as to hold to a minimum any additional cold working. The sample shall be tested in accordance with the criteria for performing the tension test prescribed by the Federal Test Method Standards No. 151.

4.4.3 Band Seating.

4.4.3.1 Destructive Method.- The rotating bands of the sample projectile shall be tested as follows: Prior to removal the diameter of the finished band shall be measured on 3 diameters 60 degrees apart and on two points along the cylindrical portion of its length (near the front and rear edge). This comprises 6 diametrical measurements. These points of measurement or orientation points shall be marked on the band and the projectile body adjacent thereto in such a manner that each measurement location can be identified. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the seat at corresponding points shall be measured. To each of the diameters of the bands seat, the sum of the corresponding band thickness shall be added and the results subtracted from the corresponding exterior diameters of the bands previously determined. The results shall indicate the total diametrical clearance between the band and the band seat.

4.4.4 Functional.

4.4.4.1 Security.- The metal parts assemblies referenced in 4.3.3.1 and 4.3.3.2 inert loaded, will be assembled into complete rounds, and fired in a gun for which the round is standard. The propellant charge will be adjusted to obtain 112 percent of the rated maximum pressure of the round in a new gun. The projectile will be fired through a target

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placed 2,000 yards from the muzzle of the gun. Photographic equipment will be set up near the target and near the muzzle of the gun in such a manner as to record the flight of the projectile prior to its impact with the target.

4.4.4.1.1 Velocity - Accuracy .- The metal parts assemblies referenced in 4.3.3.1 and 4.3.3.3 will be inert loaded and assembled into complete rounds utilizing the service charge which is standard for the round. The round will be fired to impact on a vertical target placed approximately 2,000 yards from the gun muzzle. The test rounds will be fired alternately with a group of reference rounds for comparison of velocity and accuracy at this range. Accuracy in terms of maximum spread in inches both horizontal and vertical and probable error (PE) in mils will be recorded for all rounds fired for informational purposes only (see 6.1).

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.- Packing and marking for shell H.E., 37MM, M63 shall be in accordance with drawing F7548319, and MIL-P-10025.

5.2 Data Cards.- Data card information shall be as specified in specification MIL-A-2550.

6. NOTES

6.1 Velocity-Accuracy.- Proving ground tests for velocity - accuracy are not acceptance tests, but the results will be analyzed carefully by the Ordnance Corps to determine the quality of ammunition produced. Evidence of excessive variation in any phase of the proving grounds tests will result in immediate investigation by the Ordnance Corps.

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